Ramya Koganti
Data Engineer

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PROFESSIONAL SUMMARY

- Experienced **Data Engineer** with **over 4** years designing and delivering scalable **ETL pipelines**, optimizing high-performance **data warehouses**, and implementing **cloud-native data architectures** in **healthcare** and **financial** domains to enable advanced analytics, regulatory compliance, and actionable business intelligence capabilities.
- Proficient in **Python**, **SQL**, **Spark**, and **AWS** for building real-time **data ingestion frameworks**, automating transformations, and enabling **analytics-ready datasets** supporting compliance with **HIPAA** and **SOX** regulations.
- Conceptualized and deployed **data lakes** using **Amazon S3**, **Athena**, and **Glue**, integrating **structured**, **semi-structured**, and **unstructured data** for advanced reporting, predictive analytics, and operational efficiency improvements.
- Delivered **data quality frameworks** leveraging **dbt**, **Airflow**, and **Informatica**, ensuring accurate, consistent, and complete datasets for critical healthcare analytics and high-frequency financial risk modeling workflows.
- Created **real-time streaming solutions** with **Kafka** and **Spark Streaming**, enabling low-latency analytics for fraud detection, patient monitoring, and transaction compliance in cross-domain enterprise environments.
- Collaborated cross-functionally with **data scientists**, **BI analysts**, and **cloud architects** to design end-to-end **data pipelines**, accelerating insights delivery and supporting strategic healthcare planning and investment risk analysis.
- Expertise in **data modeling**, **metadata management**, and **data governance**, ensuring interoperability across multi-source systems, optimizing query performance, and enabling secure, compliant, and high-availability data infrastructure solutions.

TECHNICAL SKILLS

Methodologies:	SDLC, Agile, Waterfall
Programming Language:	Python, SQL, R, Scala
Packages:	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, Seaborn
Visualization Tools:	Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP)
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook, IntelliJ
Cloud Platforms:	Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform
Database:	MySQL, PostgreSQL, MongoDB
Data Engineering Concept:	Apache Spark, Apache Hadoop, Apache Kafka, Apache Beam, ETL/ELT Pipeline
Other Technical Skills:	Data Lake, SSIS, SSRS, SSAS, Docker, Kubernetes, Jenkins, Terraform, Informatica, Talend, Amazon Redshift, Snowflake, Google Big Query, Data Quality and Governance, Machine Learning Algorithms, Natural Language Process, Big Data, Advance Analytics, Statistical Methods, Data Mining, Data Visualization, Data warehousing, Data transformation, A/B Testing, Git, GitHub
Operating Systems:	Windows, Linux, Mac iOS

WORK EXPERIENCE

Data Engineer | McKesson Corporation, MO-USA — **Aug 2024 – Present**

- Developed advanced data ingestion workflows using **Python**, implementing automated transformation and cleansing of patient records, enhancing ETL efficiency by **37%**, enforcing HIPAA compliance, and seamlessly integrating multi-source clinical datasets across diverse healthcare systems for improved analytics accuracy and decision-making speed.
- Engineered Validated relational data models using **SQL**, restructuring complex claims processing queries to reduce retrieval latency by **45%**, improving query performance, and enhancing real-time analytics capabilities for healthcare quality metrics, regulatory compliance reporting, and provider performance monitoring across enterprise-scale systems.
- Implemented real-time event streaming pipelines with **Apache Kafka**, integrating EHR and medical device data streams to accelerate diagnostic alerts delivery and support proactive clinical decision-making workflows in hospital networks.
- Crafted interactive healthcare performance dashboards in **Tableau**, delivering self-service BI capabilities to executives, enabling cross-departmental visibility into patient satisfaction trends, readmission rates, and operational performance KPIs.

- Provisioned scalable cloud infrastructure using **Terraform**, automating deployment of enterprise-grade data warehouses and analytics clusters, reducing provisioning time by **60%**, ensuring compliant, secure environments for large-scale healthcare analytics aligned with HIPAA and regulatory requirements.
- Optimized analytical storage solutions in **Snowflake**, implementing clustering and partitioning strategies to minimize scan costs, accelerate query performance, and enable faster, secure clinical trial data exploration for healthcare research teams.
- Containerized ETL microservices using **Docker**, ensuring consistent runtime environments across dev/test/prod, reducing deployment failures by **35%** and improving **CI/CD** pipeline efficiency for healthcare data integration processes.
- Configured machine learning models using **Python** and **TensorFlow** to predict patient readmission risks, improving predictive accuracy by **22%**, enabling proactive interventions, and integrating with EHR systems for actionable healthcare insights.
- Conducted rigorous **A/B testing** for predictive patient outcome models, validating algorithmic accuracy improvements and ensuring statistical reliability before production deployment in clinical decision support systems.
- Streamlined robust ETL workflows using **SSIS** to integrate multi-source claims and EHR datasets, ensuring high data integrity, reducing load errors by **41%**, and improving accuracy and reliability of downstream healthcare analytics processes.
- Fostered cross-functional **collaboration** between engineering, clinical, and compliance teams, aligning data initiatives with regulatory requirements, accelerating project delivery, and improving adoption of analytics-driven healthcare interventions.

Data Engineer | Infosys, India **April 2020 – July 2023**

- Constructed automated portfolio risk analysis pipelines using **Python**, streamlining complex securities data extraction, transformation, and loading, improving **ETL** efficiency by **36%**, ensuring strict Basel III compliance, and enabling accurate, timely risk assessments across diverse investment products, derivatives, and global trading platforms.
- Architected complex analytical queries using **SQL**, optimizing loan portfolio performance analysis, reducing query execution time by **42%**, and enabling rapid delivery of credit risk insights for senior financial decision-makers.
- Integrated transaction monitoring systems with **REST API** endpoints, enabling secure, real-time data exchange between core banking platforms and fraud detection modules to enhance payment security and regulatory reporting accuracy.
- Designed interactive executive dashboards in **Power BI**, visualizing key lending **KPIs**, improving decision-making efficiency by **33%**, and providing real-time financial health visibility across multiple retail and corporate banking portfolios.
- Executed serverless microservices using **AWS Lambda**, automating credit risk assessment workflows, reducing operational costs by **28%**, and enabling on-demand scalability for high-volume loan application processing.
- Trained predictive default probability models with **Scikit-learn**, applying feature engineering, hyperparameter tuning, and cross-validation to improve classification accuracy for high-risk loan detection in retail banking datasets.
- Orchestrated daily batch and streaming **ETL** workflows using **Apache Airflow**, ensuring timely processing of trade settlement, regulatory filings, and portfolio reconciliation data across distributed financial systems.
- Built large-scale **ETL** pipelines in **Informatica**, consolidating multi-source loan servicing and collateral datasets, improving data consistency for downstream analytics, and meeting stringent financial compliance requirements.
- Monitored high-frequency trading and payment system logs using **AWS CloudWatch**, configuring alerts to proactively detect anomalies and mitigate potential fraud or transaction processing delays in real-time.
- Facilitated cross-functional **collaboration** and documented analytics workflows in **Confluence**, aligning engineering, compliance, and risk management teams to accelerate project delivery, improve transparency, and ensure adherence to financial regulatory standards.

EDUCATION

Master of Science in Computer Science | University of Missouri-Kansas City, Kansas City, MO, USA
Bachelor of Technology in Computer Science | GITAM University, Hyderabad, India

CERTIFICATIONS

- **OCI Generative AI Professional - 2025**
- **OCI AI Foundations Associate – 2025**